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Executive function and verbal self-regulation in childhood: Developmental linkages between partially internalized private speech and cognitive flexibility

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ABSTRACT

Recent studies have noted that executive function and the use of self-regulatory private speech are related in childhood, and proposed that the critical leap that occurs in the development of executive function between the ages of three and six years may be due to the onset of language-based self-regulatory functions at this age. This research explored the relationship between executive function and private speech in a cross-sectional study of 81 children between four and seven years of age. The children performed an executive function task, the Dimensional Change Card Sort (DCCS), and children's use of private speech was observed during a categorization task. The results indicated that, after controlling for children's age, gender, verbal abilities, and fluid reasoning, children's use of partially internalized private speech during the categorization task was significantly related to the number of phases successfully passed on the DCCS task, which required them to switch between card sorting rules. Children who used more partially internalized private speech were more likely to pass the most challenging phase of the DCCS task that assesses the ability to flexibly use different sorting rules according to a higher-order rule. We discuss the role of verbal mediation in the development of cognitive flexibility and its implications for the design of intervention programs for children who possess deficits of executive function.

1. Introduction

A significant number of recent studies have analyzed the development of executive function (EF) during childhood (Brocki & Bohlin, 2004; Carlson & Meltzoff, 2008; Davidson, Amso, Cruess, & Diamond, 2006; Henning, Spinath, & Aschersleben, 2011). EF is defined as the skill set that allows the child to plan, monitor, and evaluate his or her performance when solving a problem (Zelazo & Frye, 1997; Zelazo, Müller, Frye, & Marcovitch, 2003). EF is an integrated construct that consists of different subcomponents: response inhibition, working memory, and attentional set shifting (Miyake et al., 2000; Zelazo, Carter, Reznick, & Frye, 1997). Each of these EF components develops gradually from the earliest years of life; however, researchers have frequently observed a critical change in the development of EF between three and six years of age (Garon, Bryson, & Smith, 2008). Despite the consistency of the evidence regarding this qualitative change in the development of EF, there is not complete agreement on possible explanations about its nature and the mechanisms that support its occurrence. Our study's main objective was to investigate the characteristics of this critical leap in the development of EF in early childhood by examining the relationship between children's performance on an EF task and children's self-regulatory private speech observed in a categorization task.

The Dimensional Change Card Sort task (DCCS; Frye, Zelazo, & Palfai, 1995; Zelazo & Frye, 1997) is a simplified version of the Wisconsin Card Sort Task (WCST; Grant & Berg, 1948) that is similar in purpose and form. The DCCS is a very useful tool to evaluate the EF of children between the ages of three and six years (Carlson, 2005; Jacques, Zelazo, Kirkham, & Semcesen, 1999; Müller, Zelazo, & Imrisek, 2005). In the standard version of the DCCS task (Zelazo, 2006), children are first asked to order a series of bivalent cards according to a rule (e.g., color), the pre-shift phase. In the second phase of the task, post-shift, the sorting rule is changed (e.g., to shape). Finally, in the most advanced phase, border phase, the children are asked to use both rules at the same time depending on the presence of a third marker on the card (e.g., a black border). In the DCCS task, children's cognitive flexibility is increasingly challenged by switching from one rule to another; first, in the post-shift phase, children have to switch to the new sorting criterion and not persevere to ordering the bivalent test cards with the rule of

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the preceding phase, and second, in the border phase, children have to flexibly switch the set of rules using a higher-order rule (Kloo, Perner, Aichhorn, & Schmidhuber, 2010; Zelazo, 2006). In a study of children between 3- and 6-year-olds, Henning et al. (2011) observed that only approximately 50% of the children between 3- and 4-year-olds passed the second phase of the DCCS (post-shift), while the majority of the 5-year-olds (80%) and 6-year-olds (92%) passed the second phase. However, 90% of the 5-year-olds and 77% of the 6-year-olds failed the third phase of the DCCS test, which requires the use of several alternative rules.

Child difficulties in the performance of the DCCS task may be due to a lack of capacity for representational flexibility (Happaney & Zelazo, 2003; Jacques et al., 1999; Müller et al., 2005). Other researchers have justified these results in terms of difficulty encountered along some of the EF dimensions, such as a problem in response inhibition (Bialystok & Martin, 2004; Carlson & Meltzoff, 2008), difficulty in maintaining the goals of the task in the working memory (Morton & Munakata, 2002; Munakata, Morton, & Yerys, 2003), or an inability to redirect the attention to an important new dimension while the previous dimension is still present ('attentional inertia'; Diamond, Kirkham, & Amso, 2002; Kirkham, Cruess, & Diamond, 2003).

The critical leap that occurs in the development of EF between the ages of three and six years may be due to the onset of language-based self-regulatory functions at this age (Zelazo & Frye, 1997; Zelazo et al., 2003). These verbal self-regulatory functions emerge when children's language becomes a tool to guide goal-directed behaviors. The verbal mediation of behavior should provide children with greater cognitive flexibility (Jacques & Zelazo, 2005; Zelazo & Frye, 1997). Some studies have found a positive association between performance in the DCCS task and children's verbal skills, an outcome that partially supports the claims about the mediating role of language in the development of EF (Carlson & Beck, 2009; Fuhs & Day, 2011; Henning et al., 2011; Müller et al., 2005).

However, the empirical studies that were devoted to test the hypothesis about the critical role of verbal self-regulatory functions in the developmental leap in EF have not provided conclusive results. These studies have focused on the use of 'verbal labeling' as a resource to test the hypothesis of verbal self-regulation in the DCCS task by asking the children to label aloud the rule that they used to sort the cards on each item. Kirkham et al. (2003), in a sample of three-year-old children, compared children's performance on a standard version of the DCCS with that on a verbal labeling condition in which children were encouraged to label the test card's sorting dimension on the first pre- and post-switch trial. Kirkham et al. found that the performance in the post-switch phase improved when the children were trained to state the label aloud. However, Müller, Zelazo, Lurye, and Liebermann (2008) failed to replicate the labeling effect in three-year-old children performing the DCCS task. Müller et al. did not find an improvement in children's performance in Kirkham et al.'s verbal labeling condition (Experiment 2). In Müller et al.'s study (Experiment 1), the labeling effect was not observed when the child was prompted to label the relevant sorting dimension either on the first trial only or on every trial.

In addition, there is no theoretical agreement among the researchers regarding why verbal labeling could improve children's execution in the DCCS task. Kirkham and Diamond (2003) proposed that the verbal labeling reduces attentional inertia, while Munakata et al. (2003) suggested that verbal labeling extends the amount of time that the rule remains in working memory. Happaney and Zelazo (2003) proposed that verbal labeling facilitates the ability to reflect on the rule system. A contradictory interpretation of the results may have been observed because the verbal labeling procedure was not adequate to assess the effect of verbal mediation on DCCS task performance. As research from the dual-task paradigm has shown, forcing children to perform a verbal task (labeling) while simultaneously performing an EF task could cause the verbal task to interfere in the performance of the EF task due to an articulatory suppression effect (Emerson & Miyake, 2003; Lidstone, Meins, & Fernyhough, 2010). On the other hand, verbal labeling the rule allows researchers to observe whether the children know the rule. The children's knowledge of the rule does not necessarily indicate that they use it in their attempts to solve the EF task (Happaney & Zelazo, 2003). Taking into account these results, we suggest that another approach to measure the effect of verbal regulation on EF development would be to examine the spontaneous task-relevant self-regulating speech used by children while they perform EF tasks (Winsler, 2009).

Several researchers have suggested that the development of EF in children may be associated with a qualitative change in the self-regulatory function of language (Fernyhough & Fradley, 2005; Jacques & Zelazo, 2005; Müller et al., 2008; Wallace, Silvers, Martin, & Kenworthy, 2009; Zelazo & Frye, 1997). Following Vygotsky (1934/1987), the self-regulatory function of language emerges between the ages of three and five years as children begin to use private speech, which is a type of speech that is not addressed toward others. It has been widely observed that children's overt private speech increases from three to five years of age, then declines in frequency as it is transformed first in partially internalized private speech (whispers and muttering), and then into verbal thought, or inner speech (Berk, 1986; Winsler, 2009; Winsler, Diaz, & Montero, 1997; Winsler & Naglieri, 2003). There are, in the extant literature, numerous studies that support these developmental trajectories in children's use of self-talk and the assumption that children's private speech is positively correlated with concurrent or future task performance (Fernyhough & Fradley, 2005; Lidstone et al., 2010; Winsler, Diaz, Atencio, McCarthy, & Chabay, 2000; Winsler, Diaz, McCarthy, Atencio, & Chabay, 1999). According to Vygotsky, the internalization of speech represents how cognitive performance begins to be verbally mediated in childhood, allowing language to become a tool for planning, controlling, and evaluating actions (Al-Namlah, Fernyhough, & Meins, 2006; Al-Namlah, Meins, & Fernyhough, 2012). Thus, verbal mediation allows children to maintain more than one rule in mind, to inhibit prepotent responses, and to keep a goal in focus; skills that are directly linked to EF (Diamond et al., 2002; Fuhs & Day, 2011). It has been claimed that more highly internalized forms of speech indicates more advanced verbal mediation, and that the rate and internalization level of children's private speech in performing a task can be representative of children's verbal self-regulatory strategies across different tasks, contexts, and timepoints (Lidstone, Meins, & Fernyhough, 2011). Winsler, De Leon, Wallace, Carlton, and Willson-Quayle (2003) found evidence for children's private speech stability across different tasks and timepoints, and consistency with children's reported behavior at school and home – children whose private speech was more partially internalized had fewer externalizing behavior problems and better social skills.

It has been proposed that the development of EF in childhood could be connected to the self-regulatory function of private speech (Fernyhough & Fradley, 2005; Winsler, 2009). Some studies have shown that overt private speech was more frequently used among children who possessed an EF deficit, such as an autism spectrum disorder or ADHD (Winsler, Abar, Feder, Schunn, & Alarcón, 2007; Winsler, Manfra, & Diaz, 2007). Other studies have reported positive relationships between the use of partially covert private speech and performance on EF tasks. In recent works, the Tower of London task (TOL; Shallice, 1982) has been used to examine the private speech used by children performing this task. The TOL task requires (a) EF-type skills (such as planning and monitoring) to place pieces in accordance with a model target and (b) the inhibition of incorrect movements. Fernyhough and Fradley (2005) evaluated children aged 5–6 years who solved a TOL task that contained items of various levels of difficulty. The results showed an increase in the frequency of externalized self-regulatory private speech use on the middle level of difficulty, a negative correlation between the use of external private speech and time needed to complete the task, and a positive association between the use of partially internalized private speech and task performance. Al-Namlah et al. (2006) showed similar results in a sample of 4–8-year-old children. The authors found a negative association between private speech production and the amount of time required to complete the TOL. In a recent study of 8–10-year-olds who performed the TOL task, Lidstone et al. (2011) found a high level of consistency in the rate of private speech and its internalization across task difficulty and time (with 11 months between observations).

Other studies have analyzed the relation between EF and inner speech according to the dual-task paradigm (Emerson & Miyake, 2003; Wallace et al., 2009). In these studies, the children completed a task of EF, such as the TOL, and simultaneously performed a secondary task involving articulatory suppression, which interfered with the use of self-regulatory speech (private or internal). Using this dual-task paradigm on 7–10-year-old children, Lidstone et al. (2010) found a significant decrease in the execution of the TOL in the articulatory suppression condition compared with the non-suppression condition. Lidstone, Fernyhough, Meins, and Whitehouse (2009) found that children with autism who possessed verbal skills were more likely than children without verbal skills or children without autism to display interference in the articulatory suppression condition when they were performing tasks that required an effort of EF. Articulatory suppression studies show that EF is reliant on self-regulatory inner speech; if children's verbal abilities were interfered with or disabled, children's abilities to plan and monitor goal-directed activities would be impaired.

The present study focuses on the development of EF in children between four and seven years of age. We selected a well-known executive task (DCCS) that assesses EF in early childhood (Zelazo, 2006; Zelazo & Frye, 1997). A significant developmental change in children's EF throughout the early childhood years has often been demonstrated using the DCCS task (Garon et al., 2008; Kloo et al., 2010; Müller et al., 2005, 2008; Zelazo et al., 2003). Children's performance on the DCCS task has been associated with other cognitive measures, such as fluid intelligence and verbal abilities (Carlson & Beck, 2009; Fuhs & Day, 2011; Henning et al., 2011; Müller et al., 2005). But studies have not explored whether children's use of private speech is associated with task performance on the DCCS; this is likely because the standard method of administering the DCCS task requires mostly social speech. Our study aimed to analyze the functional and developmental linkages between EF and self-regulatory speech in early childhood. To achieve this objective, we explored, in a cross-sectional study, the relationship of children's performance on the DCCS task and children's private speech observed while solving a separate categorization task, controlling for age, gender, fluid intelligence, and verbal abilities. Studying EF performance and private speech using the same task may improve the accuracy of analysis of the relation between the development of EF and children's private speech, because it can examine whether children use private speech when performing the EF task, for instance, coding task items for the presence of different types of speech (Winsler, 2009; Winsler, Fernyhough, McClaren, & Way, 2005). However, as observed in a pilot study (Prieto, Sánchez, & Alarcón, 2011), the formal characteristics of the DCCS task require mostly social speech. In the DCCS task, the experimenter asks the child to select in which of the "trays" a new item should be placed; thus, it is likely that the children express only social speech, in the form of answering the experimenter, during performance of the DCCS task. As research on private speech has consistently demonstrated, the best tasks for observing private speech possess an open or unstructured format, for example, building with blocks or categorization tasks (Winsler, 2009).

The current study, due to the social characteristics of the DCCS task, used a separate categorization task to observe children's private speech, and the DCCS task was used to assess the development of EF in early childhood. To the extent that children's performance on the DCCS task can be linked to children's private speech observed in a different task, recent studies have shown a strong relationship between private speech emitted during a task and performance on other tasks which measure a different cognitive ability (Al-Namlah et al., 2006, 2012). Al-Namlah and colleagues found that children's use of self-regulatory private speech during a planning task (TOL) was associated with memory performance measures such as phonological recoding (Al-Namlah et al., 2006) and autobiographical narratives (Al-Namlah et al., 2012). Assuming that rate and internalization level of child's private speech observed in the categorization task could be representative of child' private speech use on other tasks, we expect that children who used partially internalized private speech more frequently would perform better on the DCCS task.

Our hypotheses are as follows: first, a significant change in the development of EF occurs between four and seven years of age. In particular, we expect to see an increase in the use of a flexible system of rules in the DCCS task with age. From four to five years of age, the use of a sorting rule and the ability to change to another sorting rule (post-switch phase) should develop, and from six to seven years of age, the ability to alternate between rules according to a higher-order rule (border phase) should develop. Second, children were requested to complete a categorization task, where we expect to observe more frequent use of externalized private speech in the younger children and a higher amount of partially internalized private speech in the older children, indicating age-related differences in the internalization of verbal self-regulation. Third, children's performance in the EF task and children's private speech observed in the categorization task would be functionally related, with partially internalized private speech positively associated with performance on the DCCS task, after controlling for age, gender, fluid intelligence, and verbal abilities.

2. Method

2.1. Participants

Eighty-three children between four and seven years of age participated in this study, 42 of whom were boys and 41 of whom were girls. The sample consisted of pupils who attended public kindergarten and elementary schools in Western Andalusia (Spain). These schools were in middle socio-economic status environments. All of the children were born in Spain and spoke Spanish as their primary language. Two boys, who were four years of age, were eliminated from the sample because they did not pass the first phase of the DCCS test (pre-switch). The final sample, composed of 81 children, was balanced across gender and years (in months) (see Table 1).

2.2. Materials

2.2.1. Receptive vocabulary

A Spanish version of the Peabody Picture Vocabulary Test-Revised (PPVT-R; Dunn & Dunn, 1997) was administered to measure children's receptive vocabulary. The children were required to identify the picture that corresponded to a word spoken by the interviewer. Raw scores were converted into age-standardized scores. The Spanish adaptation of the PPVT-R was standardized with a representative sample of over 2500 participants from 21 provinces of Spain, in a wide range of ages

Table 1
Background variables and speech measures (utterances per minute) by age group.

Measures	Age				*F*
	4 *Mean* (*SD*)	5 *Mean* (*SD*)	6 *Mean* (*SD*)	7 *Mean* (*SD*)	
Background variables					
N (% females)	20 (50%)	22 (50%)	20 (50%)	19 (52%)	
Age in months	54.00 (3.59)	66.91 (2.92)	78.15 (3.66)	88.84 (3.76)	
Range of age	48–59	61–71	72–83	84–95	
Raven s.s.	89.55^{a} (13.56)	97.72^{ab} (14.23)	100.75^{b} (9.15)	101.42^{b} (10.13)	4.026^{*}
PPVT s.s.	99.20 (10.40)	99.13 (10.32)	102.25 (9.46)	102.00 (11.38)	0.538
WIPPSI s.s.	98.75 (10.37)	96.13 (16.25)	105.75 (8.92)	103.68 (13.00)	2.421
Speech measures					
Social speech/min	2.16^{ab} (1.31)	3.34^{a} (2.71)	1.63^{b} (0.72)	1.09^{b} (0.89)	6.363^{**}
Level 1 PS/min	0.51^{a} (0.68)	0.27^{ab} (0.41)	0.03^{b} (0.12)	0.01^{b} (.08)	7.328^{***}
Level 2 PS/min	3.52^{a} (2.91)	6.92^{b} (3.62)	2.00^{a} (2.33)	2.27^{a} (2.00)	10.646^{***}
Level 3 PS/min	0.24^{a} (0.69)	1.61^{b} (1.72)	2.08^{b} (1.53)	5.10^{c} (2.05)	28.916^{***}

Note: s.s.: standard scores; PS: private speech (utterances per minute). Means in the same row that do not share subscript letters (a–c) differ at $p < .05$ using the Tukey–Kramer method.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

(2–90-year-olds), with a split-half reliability coefficient of .91 (Dunn, Dunn, & Arribas, 2006).

2.2.2. Verbal comprehension

A Spanish version of the Similarities subscale of the Wechsler Intelligence Battery for Preschool and Primary (WPPSI-III; Wechsler, 1989) was used to assess children's verbal comprehension. For each item, the interviewer stated a pair of words that were conceptually related, and children described the relations between the words. Raw scores were converted into age-standardized scores. The Spanish adaptation of the WPPSI-III was standardized with a sample of 812 children from 4- to 7-year-olds. The split-half reliability coefficient of the verbal comprehension subscale was 0.76 (Wechsler, 2006).

2.2.3. Fluid intelligence

Children's ability to build relationships and to make analogical correlations was assessed using the Raven's Color Progressive Matrices Test (CPM; Raven, Raven, & Court, 1998). For each item, the children were presented with an image that consisted of colored geometrical shapes that were missing some small part. The children were presented with six small shapes and asked to identify which of the shapes could complete the initial picture. This intellectual capacity for comparison of shapes and analogical reasoning has been identified as fluid intelligence (Cotton et al., 2005; Raven, 2000). Raw scores were converted into age-standardized scores. The standardization of the Raven's CPM test was conducted in a Spanish sample of 1265 children between 4 and 9 years of age, with an index of test–retest reliability of 0.71, which amounts to 0.81 by the split-half procedure (Raven, Court, & Raven, 2001).

2.2.4. Executive function task

The Dimensional Change Card Sort (DCCS) task was designed to assess EF during childhood (Frye et al., 1995). In the pre-switch phase of the DCCS task, the child was required to sort a set of cards that possessed two bivalent dimensions (e.g., color/shape) according to one of the dimensions (e.g., color). During the post-switch phase, the child was required to sort the same set of cards according to the other dimension (e.g., shape). After these two phases, the child was required to switch the classification rule (color/shape) for the items based on a third marker displayed on the card (e.g., a black line on the edge of the card), the border phase. In this study, we used the standard version of the DCCS, which included the three phases described above (Zelazo, 2006). Given that the number of correct items by phase showed a left-skewed distribution, DCCS test performance was evaluated by the amount of cards correctly sorted in each of the phases, and the child's performance was classified dichotomously (success/failure) in each phase according to the following criteria: the first two phases were passed if the child ordered at least five of the six cards correctly, and the third phase was passed if the child ordered at least nine of the 12 cards correctly (Hanania & Smith, 2010; Henning et al., 2011; Müller et al., 2008). Two composite scores were computed to examine total performance in the DCCS task: (1) the sum of the correct items from all three phases (range 5–24); and (2) the number of successfully passed phases (range 1–3).

Two target cards (11 cm long × 9 cm wide) were used for the standardized DCCS task. The cards portrayed a blue horse and a red car. The two cards were placed horizontally on the table in front of each child. Two sorting trays were placed behind the target cards, ensuring that they were within reaching distance. At the beginning of the first two phases, a set of seven cards was given to the child. The cards were the same size as the target cards, and they portrayed four red horses and three blue cars. The child was required to distribute the cards appropriately. For the third phase of the test, another set of seven cards was added. These cards also portrayed four red horses and tree blue cars but included a 2 mm wide black border. The task instructions varied depending on the blocks or phases. In the preliminary (pre-switch) phase, the experimenters placed a sample card, a red horse, into a sorting tray based on its color. The remaining six cards were given to the child to place face down in the appropriate tray by color. There was no time limit to the task.

During the second phase (post-switch), a new set of seven cards was given to the child. First, the same sample card, a red horse, was placed by the experimenter according to its shape, and the child was asked to order the remaining six cards by shape. During the third phase (the border phase), all of the cards that were placed in the trays previously were collected, and two sets of cards were given to the child. One set of seven cards was similar to previous phases, and the second set of seven cards contained a black frame on the border. In this third phase, a card that did not possess black border was to be sorted according to color, and a card that possessed a black border was to be sorted based on the shape. The child was trained with two initial cards, a red horse without a border and a red horse with a black border. The child was asked to order the remaining twelve cards. The order of the sorting rules was always the same for all children, first they were requested to sort the items by the color, by shape in the second phase, and third, in the border phase, by the shape if the edge of the card was black and otherwise by the color. In each of the DCCS phases, the cards were given to each child randomly but ensuring that the same type of test card was not selected on more than two consecutive trials (Zelazo, 2006).

The instructions about the DCCS task were based on those suggested by Zelazo (2006). Along with giving the verbal instructions prior to each phase, during the rest of the task, the experimenter only asked where would be placed a new card, labeling it by the relevant dimension (e.g., “Here’s a red one. Where does this one go?”). The only difference with the standard version of the DCCS task was that at the beginning of the second phase (post-switch), the experimenter used an extra card as an example. This was done to highlight the change of sorting rule and ensure that the child understood the instructions, because empirical studies have shown that the verbal instructions regarding the end of the first phase and the beginning of the second phase are, to a certain degree, insufficient for younger children (Mack, 2007). So, after the pre-switch phase, the experimenter said to the child: “Now let’s play a new game, now we are going to sort the cards by the shape” (At this point, the experimenter showed a card with a red horse), “You see this horse should go here with the horses” (At this point, the experimenter placed the card face-down in the tray with a blue horse in front of it.).

2.2.5. Private speech

A categorization task was used to observe and record private speech. The task was based on previous literature that demonstrated the suitability of semantic tasks, such as categorizing objects, for the study of private speech (Frauenglass & Diaz, 1985; Klingler, 1986; Sánchez & Alarcón, 2006). The goal of the task was to order 25 rectangular cards. Each card was 10 cm × 10 cm. The categories of the images on the cards were food, animals, school supplies, musical instruments, and transportation. The items for each category were based on their level of typicality. For each category, two prototypical and three atypical items were chosen. All children had to sort 25 cards in five categories. They had to introduce the cards in five boxes with the name of the category attached ahead. The researcher previously sorted two extra cards as examples in each of the categories, making sure that the children understood the instructions and recognized the five categories. The same categorization task was presented to all of the children, regardless of age. In a pilot study, it was found that the difficulty level of the task was sufficiently challenging for all the children but not beyond their ability to execute it.

The children were grouped into dyads within the same age and seated at two contiguous tables. The primary reason for using a collaborative setting for the categorization task was to avoid that children could feel inhibited by the experimental situation. Our purpose was to create a task context as similar as possible to children’s classroom environment. A wide number of studies have pointed out that children’s private speech is easily observed in natural settings, such as the classroom or at home (Berk, 1986; Berk & Landau, 1993; Kraft & Berk, 1998, Winsler, Carlton, & Barry, 2000; Winsler & Diaz, 1995). In laboratory settings, where children typically are performing a task alone or either with an interviewer or with their parents, in an unfamiliar context, children might feel more inhibited. In classroom settings, children are usually engaged in tasks that require social interaction with peers. Winsler, Carlton, et al. (2000) found that preschool children observed in classroom contexts were less likely to use private speech when a teacher was present than when they were either with peers or alone. Further, there are studies that reported significant cross-context correlation between children’s private speech observed on academic tasks in the classroom with other peers and children’s private speech observed on tasks completed individually in the laboratory (Berk & Landau, 1993; Lidstone et al., 2011); thus, we assumed that there would be consistency of individual differences in children’s private speech production between the dyadic setting and an individual task.

Each child was provided with 25 cards and five small boxes in which to sort them. The boxes were lined up on the tables in front of each child, and a card with the name of the category was attached to the outer edge of each box. Two extra cards, representing typical elements of the category, were placed ahead of each box as examples of the category. The instructions provided to the children were as follows: “Here are five boxes with the name of the category and two cards next to each box. I’ll order the first cards of each group so that you know how to do it later. Do you agree? What do we have here? An elephant?” (At this point, the experimenter marked the first such card.) “What category does this one belong to? The animals, like the horse.” (At this point, the experimenter pointed to the second card and the box labeled “animals”.)

After presenting the examples, the experimenters requested that the children order the remaining 25 cards, placed in front of them, themselves. The instructions provided to the children were as follows: “You must sort those cards in each category. You may talk with each other, but everyone must do his own task. I will be waiting outside; you should call me when you are finished”. Numerous studies have shown that children engaged in tasks that require collaborating social interaction emit more private speech than in non collaborating contexts or in solitary settings (Behrend, Rosengren, & Perlmutter, 1989; Berk & Spuhl, 1995; Furrow, 1984; Goudena, 1987; Kohlberg, Yaeger, & Hjertholm, 1968), and that the amount of private speech can increase by actively encouraging children to talk to themselves (Winsler, Manfra, et al., 2007). Following Vygotsky (1934/1987), private speech emerges from children’s social interactions with others engaged in a common activity. According to this hypothesis, joint activities can be seen as facilitating the use of private speech as a tool for children’s transition from collaborative to independent task performance (Berk & Spuhl, 1995; Goudena, 1987; Winsler et al., 1997). Our proposal is that, given the social origins of private speech, dyad tasks can promote the use of both social and private speech. The current study reinforces the collaboration between the members of the dyad, and, in this way, also encourages them to talk out loud to perform the task.

The children’s speech during the categorization task was recorded, transcribed, and separated with utterances as the unit of analysis. A verbalization was considered as a discrete utterance unit if the child did not speak for at least two seconds before and after verbalization (Furrow, 1984). The coding classification system was based on existing research on children’s private speech (Winsler et al., 2005). An utterance was classified as social speech if it was clearly addressed to an interlocutor, as indicated by the use of a personal pronoun or the name of another person (e.g., “Pedro, Where is this going?”), or other signs of communicative intent, such as gestures or physical or visual contact with another person, (e.g., a child say: “This is a car.”-while looking to his or her partner and shows the card), or the utterance had the same conversational topic as the other person’s preceding utterance (e.g., a child say: “How fun is this!”, and the other child say: “Yes, this is better than being in class!”) (Winsler et al., 2005).

In contrast, an utterance was coded as private speech if it was not explicitly addressed to someone else. The private speech utterances were subclassified according to Berk’s (1986) three-level scheme as follows: Level I private speech (PS), task-irrelevant private speech (word-play/repetition, affect expression, and comments to others not present); Level II PS, task-relevant externalized private speech (over, regular volume private speech); and Level III PS, partially internalized private speech (inaudible muttering, includes whispers or silent lips movements). The card-sorting task did not possess a time limit. The percentage of each type of speech per minute was calculated to control for the differences in the amount of speech uttered due to the amount of time taken to complete the task (Lidstone et al., 2011; Winsler, Abar, et al., 2007).

2.3. Procedure

Each child performed separate tests over two different test sessions. The time between the two sessions never exceeded one week. The tests were administered by two trained experimenters in a room enabled to conduct the study at the children's school. The administration of the tests was always in the same order in each session: during the first session were administered first the vocabulary scale (PPVT-R), after the language comprehension subscale (WIPPS), and finally the fluid-reasoning tests (Raven's CPM); during the second session, the DCCS task and the categorization task were administered in this order.

The children performed the tasks individually by following the instructions of the researcher, except for the categorization task. The categorization task took place in dyads without the direct interaction with the researcher who, after given them the instructions, waited behind them at the door of the room. The object classification task was videotaped using an HDD digital camera at a distance of two meters. The subjects' voices were collected using directional microphones and were treated using a digitizing table to prevent noise. Adobe Premiere 6.0 software was used for video editing and digital sound processing.

ELAN 3.1. software was used for coding the subjects' utterances based on a speech category system. Two trained research assistants, blind to the hypothesis of the study, completed coding. To assess reliability, the two research assistants coded 20% of the transcripts used in the present analyses separately. Assigned codes for each unit of speech were compared and the omnibus Cohen's kappa for the entire speech coding system was of .85. The individual kappas for the coding speech behaviors ranged from .82 (partially internalized private speech) to .89 (social speech). The data analyses were performed using SPSS 17.

3. Results

3.1. Preliminary analyses

A 4 (Age) × 2 (Gender) MANOVA was conducted using the PPVT-R, WPPSI, and Raven's test scores as the dependent variables (see Table 1). For the standardized measure of analogical reasoning (Raven's test), there were significant differences as a function of age, $F(3, 73) = 4.02$, $p < .05$, but there were no gender differences or age × gender interaction effects. These findings are consistent with previous studies showing that, among children below six years of age, the application of the Raven's CPM test requires an additional effort to capture the interest and motivation of the children. If children have a lack of enjoyment and motivation, due to boredom or fatigue effects across the trials, the score can be well below the standard (Blair & Razza, 2007). For the standardized level of receptive vocabulary (PPVT-R), there were no significant differences between age groups, but there were differences based on gender, $F(1, 73) = 4.66$, $p < .05$. The PPVT-R was higher for girls ($M = 102.90$, $SD = 10.78$) than for boys ($M = 98.22$, $SD = 9.36$). There was no interaction effect of age × gender for receptive vocabulary. There were no significant differences in the standardized measure of verbal comprehension (WPPSI) by age, gender, nor the age × gender interaction. As there was an effect of age on the fluid intelligence measure (Raven's CPM Test), and a significant difference among gender in the receptive vocabulary test (PPVT-R), these measures are used as control measures in subsequent analyses exploring how speech and EF measures are influenced by age and gender.

3.2. Private speech measures

A 4 (Age) × 2 (Gender) MANCOVA was conducted with different types of speech utterances per minute as the dependent variables, with the measures of fluid reasoning (Raven's CPM) and verbal skills (PPVT-R and WPPSI) as the controlled covariates. The MANCOVA showed a main effect of age (see Table 1). There was no gender or age × gender interaction effect. The proportion of social speech and Level I PS rate (per minute) decreased significantly with age, $F(3, 73) = 6.36$, $p < .01$, and $F(3, 73) = 7.33$, $p < 01$, respectively. A quadratic polynomial contrast ratio ANOVA with Level II PS per minute as the dependent variable and age as the independent variable showed an inverse *U*-relation, as found in previous studies, $F(3, 77) = 4.51$, $p < .05$. Finally, the proportion of Level III PS per minute increased with age, $F(3, 73) = 28.92$, $p < .01$. Because all of the speech measures exhibit significant variance according to age, Levene's $F(7, 73) > 3$, $p < 05$, the homogeneity of covariance matrices assumption was violated. Thus, nonparametric Kruskal–Wallis Tests were performed with each type of speech as the dependent variable and age as the independent variable. The nonparametric analyses showed significant differences in all types of speech as age increased, $\chi^2(3, 81) > 15$, $p < .01$.

To examine the relation between type of speech and performance on the categorization task, which was measured as the number of items correctly classified, zero-order correlation analyses were performed. Social speech did not correlate with performance on the categorization task ($r = -.03$, $p = .77$). Level I PS was negatively related with task performance ($r = -.28$, $p < 05$), Level II PS and performance were not correlated ($r = -.01$, $p = .97$), and Level III PS was positively related with task performance ($r = .39$, $p < 01$). However, the speech measures and performance on the categorization task were not significantly correlated when age in months was controlled in partial correlation analyses.

3.3. DCCS task measures

The children's performance in each phase of the DCCS task was assessed as dichotomous (passing/falling), consistent with previous research using the DCCS (Hanania & Smith, 2010). In the pre- and post-switch phases, children were considered to pass the task if they ordered at least five of the six cards correctly, and children were considered to pass the border phase if they ordered at least nine of the 12 cards correctly. The number and percentage of children who successfully passed each phase was analyzed according to age and gender (see Table 2). In the chi-square tests, the distribution

Table 2
Numbers (and percentages) of children passing the DCCS phases and descriptive statistics of total performance scores on the DCCS task.

Variable	DCCS PS† Pass (%)	DCCS BB† Pass (%)	DCCS CI* *Mean* (*SD*)	DCCS PP* *Mean* (*SD*)
Sex				
Male ($N = 40$)	30 (75%)	8 (20%)	15.20 (5.73)	1.95 (0.67)
Female ($N = 41$)	34 (82%)	12 (29%)	16.87 (5.27)	2.12 (0.67)
Age				
4 ($N = 20$)	12 (60%)	1 (5%)	12.25^{a} (5.24)	1.65^{a} (0.58)
5 ($N = 22$)	15 (68%)	3 (13%)	14.68^{ab} (6.27)	1.81^{ab} (0.66)
6 ($N = 20$)	19 (95%)	6 (30%)	17.65^{bc} (3.52)	2.25^{bc} (0.55)
7 ($N = 19$)	18 (94%)	10 (52%)	19.94^{c} (3.30)	2.47^{c} (0.61)

Note: PS: post-switch; BB: black border; CI: number of correct items; PP: number of passed phases. Means in the same column that do not share subscript letters (a–c) differ at $p < .05$ using the Tukey–Kramer method.

† Age-group differences were significant (χ^2, $p < .01$).

* Age-group differences were significant (F, $p < .01$).

of children who passed and those who did not pass the post-switch phase by age groups deviated significantly from the expected null distribution, χ^2 (3, 81) = 11.83, $p < .01$. Thus, the 4- and 5-year-old children frequently failed when they had to switch sorting rules (40% and 32%, respectively), while most of the 6- and 7-year-old children performed successfully (95% and 94.7%, respectively). Performance in the last phase of the DCCS task (border) improved significantly with age, χ^2 (3, 81) = 13.89, $p < .01$, with 5%, 13.6%, 30% and 52.6% of the 4-, 5-, 6- and 7-year-old children successfully passing, respectively. There were no significant differences according to gender in the distribution of children who passed and those who did not pass the post-switch and border phases.

To examine total performance in the DCCS task, two composite scores were computed: (1) the sum of the correct items over all three phases (range 5–24), and (2) the number of successfully passed phases (range 1–3). Separate 4 (Age) × 2 (Gender) ANCOVAs were conducted using the total composite scores as the dependent variables, with the measures of fluid reasoning (Raven's CPM) and verbal skills (PPVT-R and WPPSI) as covariates. The results showed significant differences by age group (see Table 2) in the total number of items correctly sorted, F (3, 7) = 6.54, $p < .01$, and the number of phases overcome, F (3, 70) = 5.16, $p < 01$. However, there were no differences between boys and girls, and no age × gender interaction effects for either of the composite measures of DCCS task performance. As the assumption of homogeneity of variance among groups was violated for the performance measures in the DCCS task, Levene's F (7, 73) > 4, $p < 05$, nonparametric Kruskal–Wallis Tests were performed with each of the composite performance scores as dependent variables and age as the independent variable. These nonparametric tests showed significant differences between age groups in the number of items correctly sorted, χ^2 (3, 81) = 24.29, $p < .001$, and the number of phases successfully passed, χ^2 (3, 81) = 18.62, $p < .001$.

3.4. Executive function and private speech relational analysis

Table 3 displays the zero-order correlations between the DCCS task measures, the rates of utterances per minute, cognitive abilities, and age in months. The total performance measures in the DCCS task, the number of correct items, and the number of phases successfully passed correlated negatively with the rate of Level II PS per minute, r (81) = −.22, $p < 05$, and r (81) = −.25, $p < .05$, respectively. The number of correct items and number of phases successfully passed were positively associated with the rate Level III PS per minute, r (81) = .50, $p < .01$, and r (81) = .53, $p < .01$, respectively. As age of the children was positively correlated with the number of correct items and with the number of phases successfully passed, age in months was controlled in a partial correlation analysis. However, after controlling for age (in months), only the number of phases successfully passed and the rate of Level III PS per minute were significantly positively associated, r (78) = .25, $p < .05$ (see Table 3).

To find out whether the EF performance measures are associated with the self-regulatory private speech utterances, after controlling for the effects of the background variables, regressions analyses were computed with EF measures as the dependent variables and age in months, gender, verbal abilities tests, fluid intelligence, and task-relevant private speech categories as independent variables. First, to test the hypothesis that private speech measures would be associated with successful performance in the post-switch and border phases of the DCCS task, which required that the children changed the sorting rule, two hierarchical logistic regression analyses were conducted using the dichotomous performance measure (success/failure) on each phase of the DCCS task as the criterion variables (see Table 4). The first model showed that age accounted for a significant amount of the variance in passing the DCCS post-switch phase, χ^2 (2, 81) = 15.33, $p < .01$, Nagelkerke's $R^2 = .27$. After controlling for age and gender, only the PPVT-R standardized score, entered on the second block, accounted for a significant amount of the variance on passing the DCCS post-switch phase, χ^2 (3, 81) = 14.13, $p < .01$, Nagelkerke's $R^2 = .47$. The analogical reasoning (Raven's CPM) and WPPSI scores, entered on the second block, and the rates of Level II PS and Level III PS per minute, entered on the third block, did not account for a significant amount of the variance in passing the DCCS post-switch phase. In the second logistic regression, age (entered in the first block) was significantly associated to successfully pass the DCCS border phase, χ^2 (2, 81) = 21.28, $p < .01$, Nagelkerke's $R^2 = .34$. The standardized measures of Raven's CPM, PPVT-R, and WPPSI tests, entered on the second block, did not account for a significant amount of the variance in passing the DCCS border phase. Level III PS per minute, entered in the third block, accounted for a significant amount of the variance in passing the DCCS border phase, χ^2 (1, 81) = 8.53, $p < .05$, Nagelkerke's $R^2 = .47$. Table 4 shows the results of the models from both analyses, including all variables entered.

Second, to investigate which type of private speech may be related to overall performance on the EF task, two hierarchical linear regression models were conducted with overall amount of correct items on the DCCS task and the number of successfully passed phases as the dependent variables. Age (in months), gender, rates of task-relevant private speech per minute, standardized scores on the Raven's CPM, PPVT-R, and WPPSI were used as independent variables (see Table 5). The first linear regression model showed that age, entered in the first block, was significantly associated with the amount of correct items on the DCCS task, F (2, 78) = 21.44, $p < .01$, $R^2 = .35$. After controlling for the effects of age

Table 3
Zero order and partial correlations among measures ($N = 81$).[a]

Measures	1	2	3	4	5	6	7	8	9	10
1. Age (months)	1									
2. Raven s.s.	.33**	1								
3. PPVT s.s	.11	32* (.30)**	1							
4. WIPPS s.s.	.17	.15 (.09)	.36** (.35)**	1						
5. DCCS CI	.57***	.33** (.18)	.35** (.36)**	.26* (.20)	1					
6. DCCS PP	.54***	.27* (.11)	.30* (.29)**	.23* (.17)	.90*** (.85)***	1				
7. Social speech/min	−.27	.02 (.12)	.08 (.11)	−.18 (−.14)	−.17 (−.02)	−.17 (−.04)	1			
8. Level I PS/min	−.41***	−.02 (.14)	−.07 (−.03)	−.03 (.05)	−.21 (.03)	−.21 (.01)	.08 (−.04)	1		
9. Level II PS/min	−.23*	−.05 (.03)	−.10 (−.07)	−.21 (−.17)	−.22* (−.11)	−.25* (−.15)	.54*** (.51)***	.03 (−.07)	1	
10. Level III PS/min	.69***	.21 (−.02)	.10 (.03)	.16 (.05)	.50*** (.18)	.53*** (.25)*	−.34** (−.23)*	−.32** (−.05)	−.11 (.06)	1

Note: s.s.: standard scores; CI: number of correct items; PP: number of passed phases; PS: private speech (utterances per minute).
[a] Partial correlations after controlling for age in months are reported in parentheses.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

Table 4
Hierarchical logistic regression analyses for variables predicting performance on the phases of DCCS task.

Variable	DCCS post-switch				DCCS border			
	B	S.E. B	Wald	ΔR^2	B	S.E. B	Wald	ΔR^2
Model 1				.27***				.34***
Age (months)	.09	.02	10.87**		.11	.03	13.64***	
Gender (male = 1)	−.65	.61	1.13		−.68	.60	1.26	
Model 2				.20***				.01
Age (months)	.10	.03	8.37**		.12	.03	12.18***	
Gender (male = 1)	−.77	.73	1.14		−.58	.65	.80	
Raven s.s.	.03	.03	.95		−.01	.03	.12	
PPVT s.s.	.11	.05	5.10*		.03	.03	.55	
WIPPSI s.s.	.04	.03	1.70		−.01	.03	.27	
Model 3				.01				.12*
Age (months)	.10	.04	4.97*		.05	.04	1.89	
Gender (male = 1)	−.73	.73	1.01		−.78	.73	1.15	
Raven s.s.	.03	.03	.93		−.01	.03	.08	
PPVT s.s.	.11	.05	4.90*		.03	.04	.55	
WIPPSI s.s.	.04	.03	1.43		−.03	.03	.68	
Level II PS/min	−.06	.10	.39		−.19	.14	1.68	
Level III PS/min	.04	.22	.04		.47	.18	6.83**	
Total R^2	.48				.47			
χ^2 for model 3	29.74***			30.45***				

Note: s.s.: standard scores; PS: private speech (utterances per minute). The R-squared reported are Nagelkerke's R^2.
* $p < .05$.
** $p < .01$.
*** $p < .001$.

and gender, the PPVT-R score, entered in the second block, was the only variable that accounted for a significant amount of the variance in number of correct items on the DCCS task, $F(5, 75) = 11.47$, $p < .01$, $R^2 = .43$ [$\Delta R^2 = .08$, $F(3, 75) = 3.46$, $p = .02$, Durbin–Watson = 1.532]. The second linear regression analysis showed that age, entered in the first block, was significantly related to the number of successfully passed phases of the DCCS task, $F(2, 78) = 17.44$, $p < .01$, $R^2 = .31$. After controlling for age, the variables gender, Raven's CPM, PPVT-R, and WPPSI scores, entered in the second block, did not account for a significant amount of the variance in the performance measure on the DCCS task. In the third block, age in months and the rate of Level III PS per minute were the only variables that accounted for a significant amount of the variance in the number of successfully passed phases of the DCCS task, $F(6, 74) = 7.37$, $p < .01$, $R^2 = .41$ [$\Delta R^2 = .05$, $F(1, 74) = 3.29$, $p = .043$, Durbin–Watson = 1.488].

4. Discussion

The different phases of the DCCS task require increasing levels of rule complexity (Happaney & Zelazo, 2003; Zelazo et al., 2003). The analysis of children's performance in the DCCS task showed, as hypothesized, that the children were able to switch from one sorting rule to another (post-shift phase) at approximately four or five years of age, although the children were not able to use a higher-order rule that allowed them to flexibly change the rule continuously (border phase) until they were six or seven years of age. These results are consistent with those found in previous research using the same or other versions of the DCCS task (Kloo et al., 2010; Müller et al., 2005, 2008). However, most of the previous research focused on the age at which children successfully pass the post-switch phase, and few studies have provided conclusive data for

Table 5
Multiple linear regression analyses for variables predicting total performance on DCCS task.

Variable	DCCS correct items				DCCS passed phases			
	B	SE B	β	ΔR^2	B	SE B	β	ΔR^2
Model 1				.35***				.31***
Age (months)	.24	.04	.58***		.03	.00	.54***	
Gender (male = 1)	−1.77	1.00	−.16		−.18	.13	−.14	
Model 2				.08*				.05
Age (months)	.21	.04	.51***		.02	.00	.50***	
Gender (male = 1)	−1.13	.98	−.10		−.11	.13	−.09	
Raven s.s.	.03	.04	.07		.00	.00	.03	
PPVT s.s.	.12	.05	.22*		.01	.01	.19	
WIPPSI s.s.	.03	.04	.08		.00	.00	.06	
Model 3				.02				.05*
Age (months)	.15	.05	.36**		.01	.01	.27*	
Gender (male = 1)	−1.05	.99	−.10		−.10	.13	−.07	
Raven s.s.	.03	.04	.08		.00	.00	.04	
PPVT s.s.	.12	.05	.22*		.01	.01	.19	
WIPPSI s.s.	.02	.04	.06		.00	.00	.04	
Level II PS/min	−.10	.15	−.06		−.02	.02	−.11	
Level III PS/min	.48	.28	.20		.08	.04	.29*	
Total R^2	.45				.41			
F for model 3	8.77***				7.37***			

Note: s.s.: standard scores; PS: private speech (utterances per minute).
* $p < .05$.
** $p < .01$.
*** $p < .001$.

the border phase. Recently, Henning et al. (2011) noted that only a small percentage of children who were six years of age (23%) were able to overcome the border phase of the DCCS task. Our study provides novel data on the age at which children mostly passed the border phase: 30% at 6 years and 52% at 7 years.

The results of the types of speech used by children while they performed the categorization task showed significant differences in the rate of self-regulatory private speech per minute between 4- and 7-year-olds. The younger children, the 4- and 5-year-olds, most frequently used social speech and Level II PS, while the older children, the 6- and 7-year-olds, more frequently used Level III PS. These results indicate that in early childhood, confirming our second hypothesis, children's verbal self-regulation is promoted first by the use of Level II PS and later replaced by the use of Level III PS, as has been observed in previous studies of private speech for a variety of task, contexts, and cultural groups (Al-Namlah et al., 2006; Lidstone et al., 2011).

Our results on the development of EF and self-regulating private speech were similar to those found in previous research (Fernyhough & Fradley, 2005; Winsler et al., 1997); however, the principal aim of the study was to explore the relationship between these two constructs. As hypothesized, our results suggest that the EF performance in childhood is linked to the internalization of private speech. EF and private speech share cognitive functions (behavior self-regulation) and developmental patterns. To test this hypothesis, we conducted several logistic and linear regression models with performance of the DCCS task as criterion variables and age, gender, verbal abilities, fluid intelligence, and rates of task-relevant private speech per minute measures as the independent variables.

The logistic regression models used the categorical performance of children in each of the phases (passing/failing). The results from these models showed that age (in months) and the verbal ability score (PPVT-R) were the main factors associated with the children's performance in the post-switch phase. In the border phase, age was a significant factor; however, even after controlling for age, gender, fluid intelligence, and verbal abilities measures, the rate of Level III PS per minute was significantly associated with children's successful performance. The linear regression models evaluated the effect of Level III PS, among other variables, on the total execution of the DCCS task. The results showed that age (in months) and the verbal ability score (PPVT-R) accounted for the greatest amount of variance in the number of correct items on the DCCS task. However, after controlling for age, gender, verbal abilities, and fluid intelligence, the rate of Level III PS per minute was significantly associated with the number of phases successfully passed. These findings suggest that, in terms of quantitative changes in the performance of the DCCS task, age and verbal ability are positively related with the number of correct items, and the amount of partially internalized private speech utterances per minute is significantly associated with the qualitative change between phases. Independently of the background variables, the children who used more frequent partially internalized private speech were more likely to correctly switch between the sorting rules, and, therefore, they were more likely to pass the most challenging phase of the EF task, the border phase, which required them to flexibly switch between two card sorting rules according to a higher-order rule.

In summary, the present study indicates that the development of the EF is staggered. The ability to change a rule, post-shift phase, occurs at approximately four to five years of age, while the flexible use of various rules, border phase, occurs by six to seven years of age. Furthermore, our study shows that the age-related differences in the internalization of self-regulatory private speech, observed by the lower use of externalized private speech and the higher rate of partially internalized private speech in the older children, occurs in the same staggered pattern as the development of EF. We suggest that these two cognitive development patterns are linked beyond the chronological calendar and that the internalization of private speech may facilitate the increased flexibility in rule sorting. This increase may be shown by the relation between children's use of partially internalized private speech and performance on the DCCS task, especially during the border phase that required the most cognitive flexibility.

This study possesses certain limitations, such as the assessment of EF based only on the standard DCCS task. It would be appropriate to replicate these results using other versions of the DCCS or other EF tasks (Carlson, 2005; Carlson & Meltzoff, 2008; Müller, Steven Dick, Gela, Overton, & Zelazo, 2006). Another limitation of our study was that the data for the relationship between private speech and EF were drawn from separate tasks, the DCCS task and the categorization task. It may be appropriate to assess the use of private speech directly on the execution of the DCCS task; however, most of the children's utterances during the DCCS task were purely social because the experimenter asked them to order each new item explicitly. The main reason that the DCCS task was used in this study was because our aim was to explore the qualitative leap in the development of EF in the early childhood years, as indicated by the fact that children of different ages get overcome a few phases of the DCCS task but not others more cognitive challenging. We found that the performance on the DCCS task was associated to the level of internalization of the children's private speech, and this relation was independent of children's age, gender, verbal abilities, and fluid intelligence. Previous studies have employed EF tasks that were best suited to monitor the use of private speech, such as the TOL, in which children are required to solve a manipulative problem. These studies have observed similar trends in the patterns of internalization of private speech and the increased ability of children's cognitive flexibility (Al-Namlah et al., 2006; Fernyhough & Fradley, 2005; Lidstone et al., 2011).

In conclusion, the current study replicates the findings that EF is related to verbal ability and internalization of private speech, which may indicate that verbal mediation plays a role in the development of cognitive flexibility (Fuhs & Day, 2011; Jacques & Zelazo, 2005; Zelazo & Frye, 1997). One possible line of future research could be to observe the use of private speech in a computerized version of the DCCS task under different experimental conditions (Müller et al., 2006). Thus, if instructions were audibly played by the computer, without an experimenter present, and the child was requested to order the items displayed in the screen, the child's utterances would consist entirely of purely private speech, making it possible to test whether similar patterns of development of EF and children's use of private speech could be observed in the standard and other versions of the DCCS task. In addition, our finding of a positive association between partially internalized private speech and EF performance suggests that future research could explore whether the private speech facilitates the development of EF. Some longitudinal studies have shown that private speech is strongly correlated with future task success rather than concurrent success (Behrend et al., 1989; Bivens & Berk, 1990; Winsler et al., 1997). The current study analyzed the relationship between the level of internalization of private speech and the EF in a cross-sectional study, but a challenge for future research is to determine whether developmental patterns of private speech could be a predictor of EF performance over time. The design of training programs promoting the internalization of verbal self-regulation may be appropriate for children who possess deficits in EF, although it would be necessary to test this hypothesis using longitudinal studies that include pre- and post-test measures (Barnett et al., 2008; Diamond, Barnett, Thomas, & Munro, 2007; Diamond & Lee, 2011; Ozonoff & McEvoy, 1994). From a Vygotskian perspective, such internalization of private speech represents the transition from the external social regulation of behavior to the use of speech as a tool to regulate and guide own behavior. If private speech and EF are developmentally related, teachers could encouraged children to talk to themselves during EF and other tasks and scaffold their performance by providing them with instructions that children can repeat themselves (Diamond et al., 2007; Diamond & Lee, 2011; Winsler, 2009).

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